

Section 7.1: Introduction

Def. A **statistic** is a function of

1.

&

2.

Some examples:

1. Sample mean: $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$

2.

3.

Def. The probability distribution of a statistic is called a _____.

Example 1. A fair six-sided die is rolled three times. Let Y_1 , Y_2 , and Y_3 denote the outcome of rolls 1, 2, and 3, respectively.

(a) Find the probability distribution, mean, and variance for one roll.

- (b) Find the mean of $\bar{Y}$, $\mu_{\bar{Y}}$.
- (c) Find the standard deviation of $\bar{Y}$, $\sigma_{\bar{Y}}$.
- (d) How could we find the sampling distribution of $\bar{Y}$?

Question. How can we get an idea about the shape of the sampling distribution of $\bar{Y}$ without doing all of these calculations?

Section 7.2: Sampling Distributions Related to the Normal Distribution

Important Abbreviation. *i.i.d.* means _____.

Theorem 7.1. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then,

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \sim$$

Proof. Recall the following theorem from Handout 25 (MAT 310): If $Y_1, Y_2, \dots, Y_n$ are independent Normal RVs such that $Y_i \sim N(\mu_i, \sigma_i)$ and $X = \sum_{i=1}^n a_i Y_i$, then

$$X \sim$$

Fact 1. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} = \sqrt{n} \left(\frac{\bar{Y} - \mu}{\sigma} \right)$. Then,

$$Z \sim$$

Example 2. Suppose that light bulbs have a length of life that is normally distributed with mean μ and standard deviation $\sigma = 40$ hours. Suppose we take a random sample of 16 light bulbs and measure their lengths of life.

- (a) Find the probability that the sample mean will be within 10 hours of the population mean.

- (b) Suppose we would like the sample mean to be within 10 hours of the population mean with probability 0.90. How many light bulbs must we sample in order to ensure this degree of accuracy?

Fact 2. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $Z_i = \frac{Y_i - \mu}{\sigma}$. Then,

$$Z_i^2 \sim$$

Theorem 7.2. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $Z_i = \frac{Y_i - \mu}{\sigma}$. Then,

$$\sum_{i=1}^n Z_i^2$$

Proof.

Question. How can we use software to find probabilities for $Y \sim \chi_\nu^2$?

- $P(Y \leq y) =$

-

Question. How can we find the $(100 \cdot p)^{th}$ percentile, ϕ_p , of $Y \sim \chi_\nu^2$?

- $\phi_{0.95}$ of a χ_6^2 RV:
- $\phi_{0.10}$ of a χ_{21}^2 RV:

Example 3. Suppose $Z_1, Z_2, \dots, Z_{10} \stackrel{i.i.d.}{\sim} N(0, 1)$.

- (a) Find b such that $P\left(\sum_{i=1}^{10} Z_i^2 \geq b\right) = 0.95$.

- (b) Find b such that $P\left(\sum_{i=1}^{10} Z_i^2 < b\right) = 0.10$.

Fact 3. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then, $\bar{Y}$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ are independent.

Theorem 7.3. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then,

$$\frac{(n-1)S^2}{\sigma^2} \sim$$

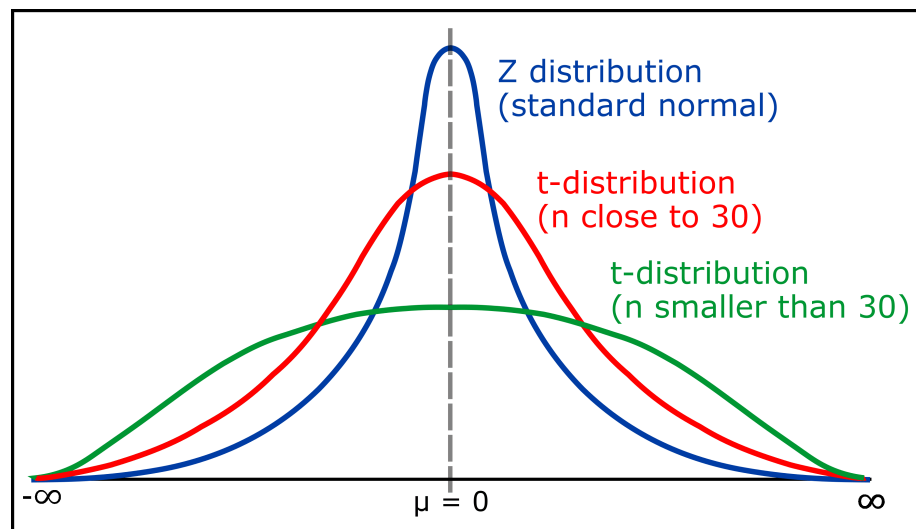
Proof.

Example 2 cont'd. Suppose that light bulbs have a length of life that is normally distributed with mean μ and standard deviation $\sigma = 40$ hours. Suppose we take a random sample of 16 light bulbs and measure their lengths of life.

- (c) Find an interval that will contain S^2 with probability 0.95.

Remark. Theorem 7.1 provides the basis for inference-making procedures about μ for a Normal population with standard deviation, σ .

Question. What if σ is unknown?



Fact 4. Suppose $Z \sim N(0, 1)$ and $W \sim \chi_\nu^2$. Then, if Z and W are independent,

$$T = \frac{Z}{\sqrt{W/\nu}} \sim$$

Fact 5. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $T = \frac{\bar{Y} - \mu}{S/\sqrt{n}} = \sqrt{n} \left(\frac{\bar{Y} - \mu}{S} \right)$. Then,

$$T \sim$$

Proof.

Question. How can we use software to find probabilities for $Y \sim t_{n-1}$?

- $P(Y \leq y) =$

Question. How can we find the $(100 \cdot p)^{th}$ percentile, ϕ_p , of $Y \sim t_{n-1}$?

- $\phi_{0.95}$ of a t_6 RV:
- $\phi_{0.10}$ of a t_{21} RV:

Example 4. Suppose it is known that the GPAs of freshmen are normally distributed with unknown mean and variance, μ and σ^2 , respectively. We take a random sample of 8 freshmen. Find the probability that $\bar{Y}$ will be within $3S/\sqrt{8}$ of the population mean, μ .

Fact 6. Let $W_1 \sim \chi_{\nu_1}^2$ and $W_2 \sim \chi_{\nu_2}^2$ be independent RVs. Then,

$$F = \frac{W_1/\nu_1}{W_2/\nu_2} \sim$$

Question. Consider two independent random samples from normal distributions. One sample is size n_1 with variance σ_1^2 ; the other is size n_2 with variance σ_2^2 . Let S_1^2 and S_2^2 represent the estimators of σ_1^2 and σ_2^2 , respectively. What does the above definition imply about the distribution of $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$?

Question. How can we use software to find probabilities for $Y \sim F_{\nu_1, \nu_2}$?

- $P(Y \leq y) =$

Question. How can we find the $(100 \cdot p)^{th}$ percentile, ϕ_p , of $Y \sim F_{\nu_1, \nu_2}$?

- $\phi_{0.95}$ of a $F_{4,7}$ RV:
- $\phi_{0.995}$ of a $F_{2,9}$ RV:

Example 5. Suppose we have independent random samples from two normal populations with $n_1 = 16$, $\sigma_1^2 = 10$, $n_2 = 25$, and $\sigma_2^2 = 15$.

- (a) Find the number b such that $P\left(\frac{S_1^2}{S_2^2} \leq b\right) = 0.99$.

- (b) Find the number c such that $P\left(\frac{S_1^2}{S_2^2} \geq c\right) = 0.99$.

Section 7.3: The Central Limit Theorem

Question. What is the distribution of $\bar{Y}$ for i.i.d. Y_i that are not normal? Let's look at the following applet: https://digitalfirst.bfwpub.com/stats_applet/stats_applet_3_cltmean.html.

Theorem 7.4: Central Limit Theorem (CLT). Let $Y_1, Y_2, \dots, Y_n$ be i.i.d. RVs with $E(Y_i) = \mu$ and $V(Y_i) = \sigma^2 < \infty$. Then, as $n \rightarrow \infty$,

$$\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \text{ converges in distribution to } \underline{\hspace{2cm}}.$$

In other words, $\bar{Y}$ is asymptotically normal with mean $\underline{\hspace{1cm}}$ and variance $\underline{\hspace{1cm}}$.

Remark. Regardless of the distribution of the i.i.d. Y_i , the CLT implies that $\bar{Y}$ is approximately normal for large n . But how large is “large enough”?

Example 6. The weights of adult female elk are i.i.d. with a mean of 490 pounds and a standard deviation of 45 pounds. Compute the probability that the average weight of 100 female elk is above 500 pounds.

Remark. Review the CLT examples from MAT 310!

Other Important Distribution Results

Fact 7. Suppose $Y_1, \dots, Y_n \stackrel{\text{indep.}}{\sim} \text{Gamma}(\alpha_i, \beta)$. Let $U = \sum_{i=1}^n Y_i$. Then,

$$U \sim$$

Proof.

Fact 8. Suppose $Y \sim \text{Gamma}(\alpha = \frac{n}{2}, \beta)$, where n is a positive integer. Let $W = \frac{2Y}{\beta}$. Then,

$$W \sim$$